INCH-POUND

MIL-M-28787/225B
AMENDMENT 1
30 July 1993

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SWITCH, S/D, MULTISPEED, ANALOG, KEY CODE HUH

This amendment forms a part of Military Specification MIL-M-28787/225B(EC), dated 12 October 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 12

TABLE III, Test condition B4, Unit column: Delete "Vp-p" and substitute "mVp-p".

PAGE 13

FIGURE 5: Delete and substitute new figure 5 as printed on page 3 of this amendment.

PAGE 14

FIGURE 6: Delete "waveform at pin 34" (two places) and substitute "waveform at pin 35" (two places).

PAGE 16

FIGURE 7, NOTE 2: Delete in its entirety and substitute "2. R2 = R3.".

PAGE 17

FIGURE 8: Delete and substitute new figure 8 as printed on page 4 of this amendment.

PAGE 19

FIGURE 9, NOTE 2: Delete "R2 = 45.3 Ohm" and substitute "R2 = 45.3 KOhm".

PAGE 28

FIGURE 15: Delete and substitute new figure 15 as printed on page 5 of this amendment.

PAGE 29

FIGURE 16, top left hand side of page, ITEM J02 OUTPUT PULSE: Delete "250 ± 20 ms" and substitute "125 ± 10 ms" on both high and low pulse widths.

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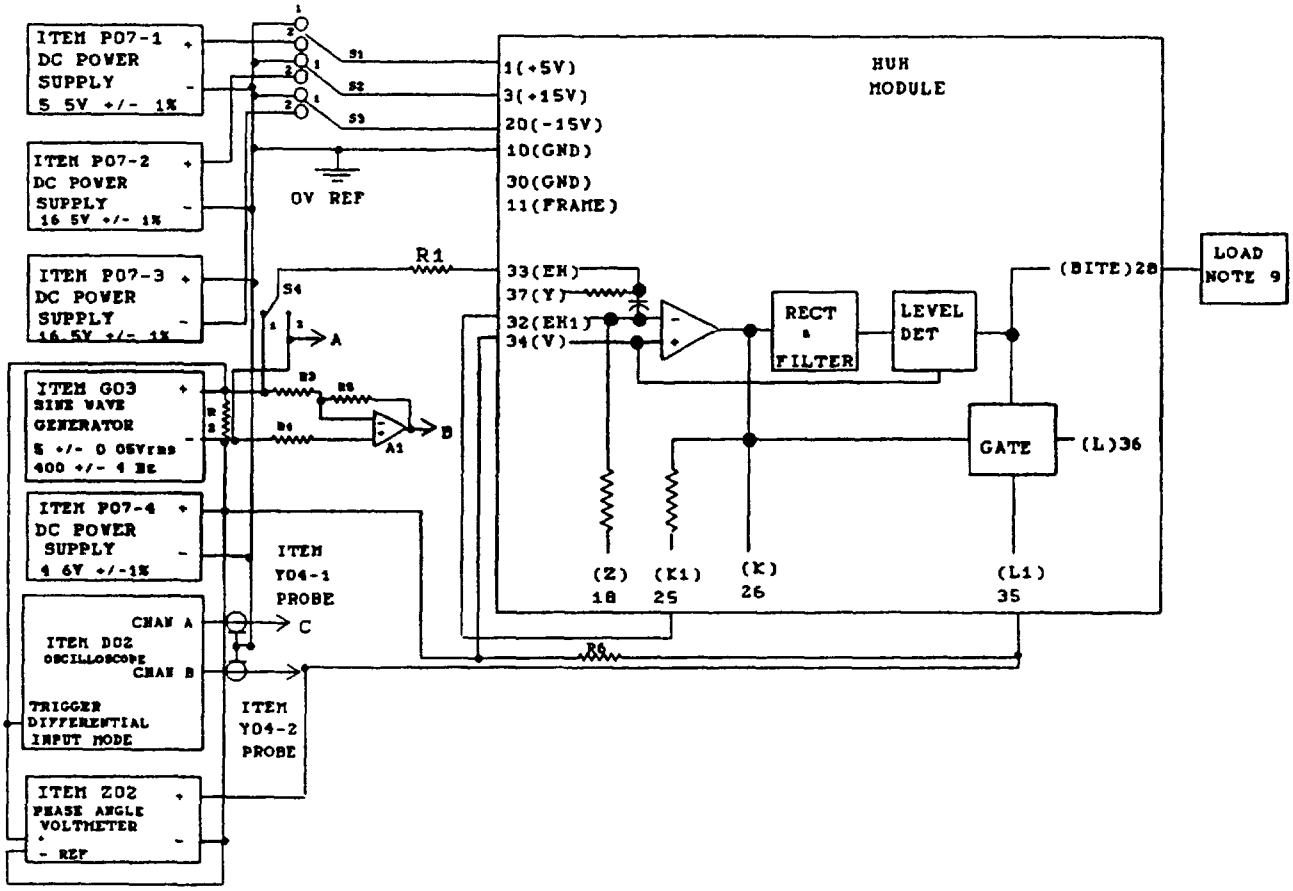
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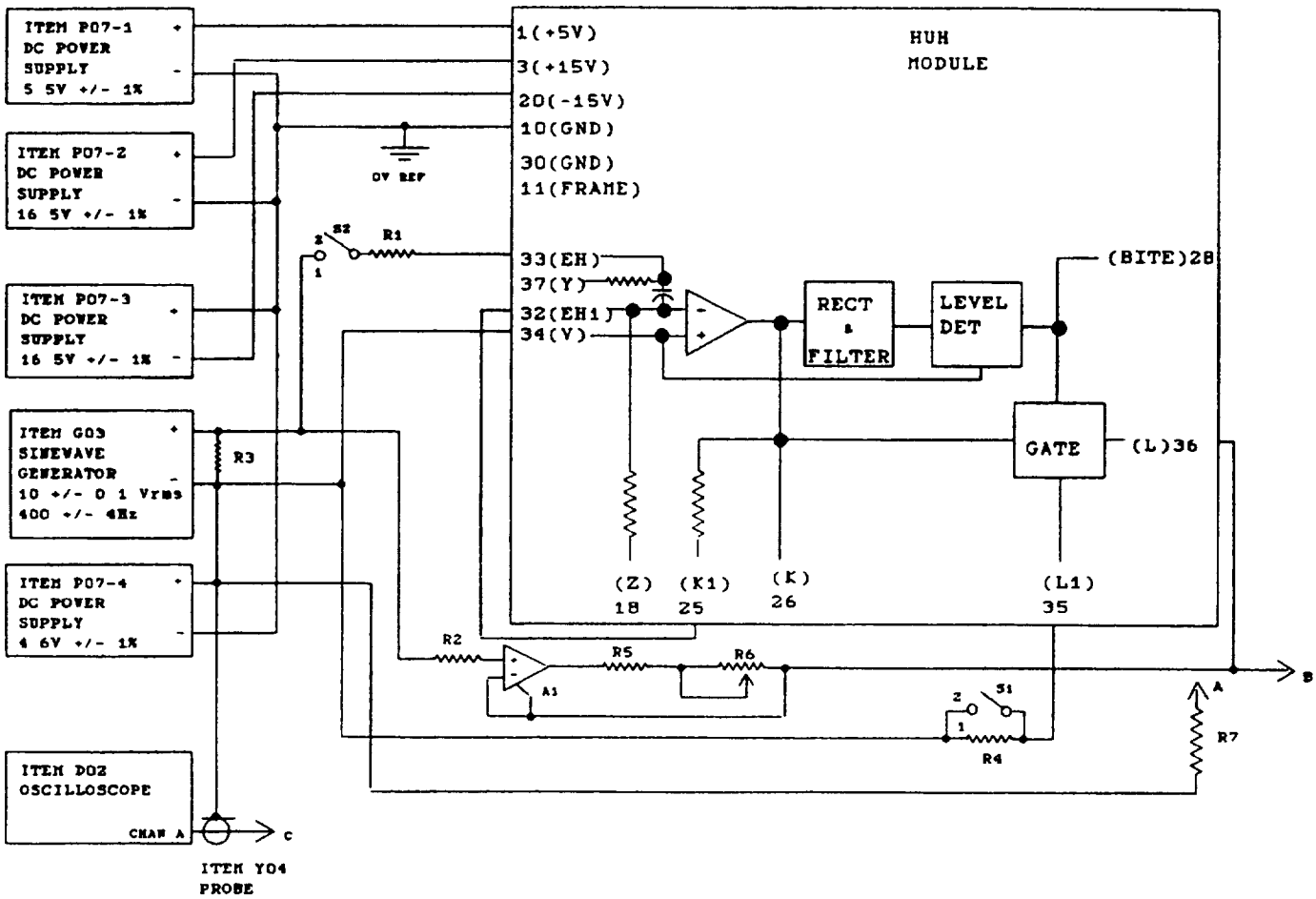
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NOTES:
1. R1 = 45.3 kOhm ± 1%, 1/8 W.
2. R2 = 600 Ohm ± 10%, 1/8 W.
3. R3 = 10 kOhm ± 1%, 1/8 W.
4. R4 = 5 kOhm ± 1%, 1/8 W.
5. R5 = 10 kOhm ± 1%, 1/8 W.
6. R6 = 20 kOhm ± 1%, 1/8 W.
7. S1-S4 = SPDT switch.
8. A1 = Fairchild type μA741 OpAmp, or equivalent.
9. Load BITE (pin 28) per figure 10, note 6.
10. Switching sequence:
    (a) Position switches S1-S4 to position 1; operate switch S1 to position 2, S2 to position 2, then S3 to position 2; perform test conditions B1, B2, and B3.
    (b) Reposition switches S1-S3 to position 1; operate switch S2 to position 2, S3 to position 2, then S1 to position 2; perform test conditions B4, B5, and B6.
    (c) Reposition switches S1-S3 to position 1; operate switch S3 to position 2, S1 to position 2, then S2 to position 2; perform test conditions B7, B8, B9, and B10.
11. For test conditions B1, B2, B5-B8, the measurements are taken above or below analog ground which is = 4.6 V above the 0 V REF (see figure 6).

*FIGURE 5. Hot socket test circuit for table III.

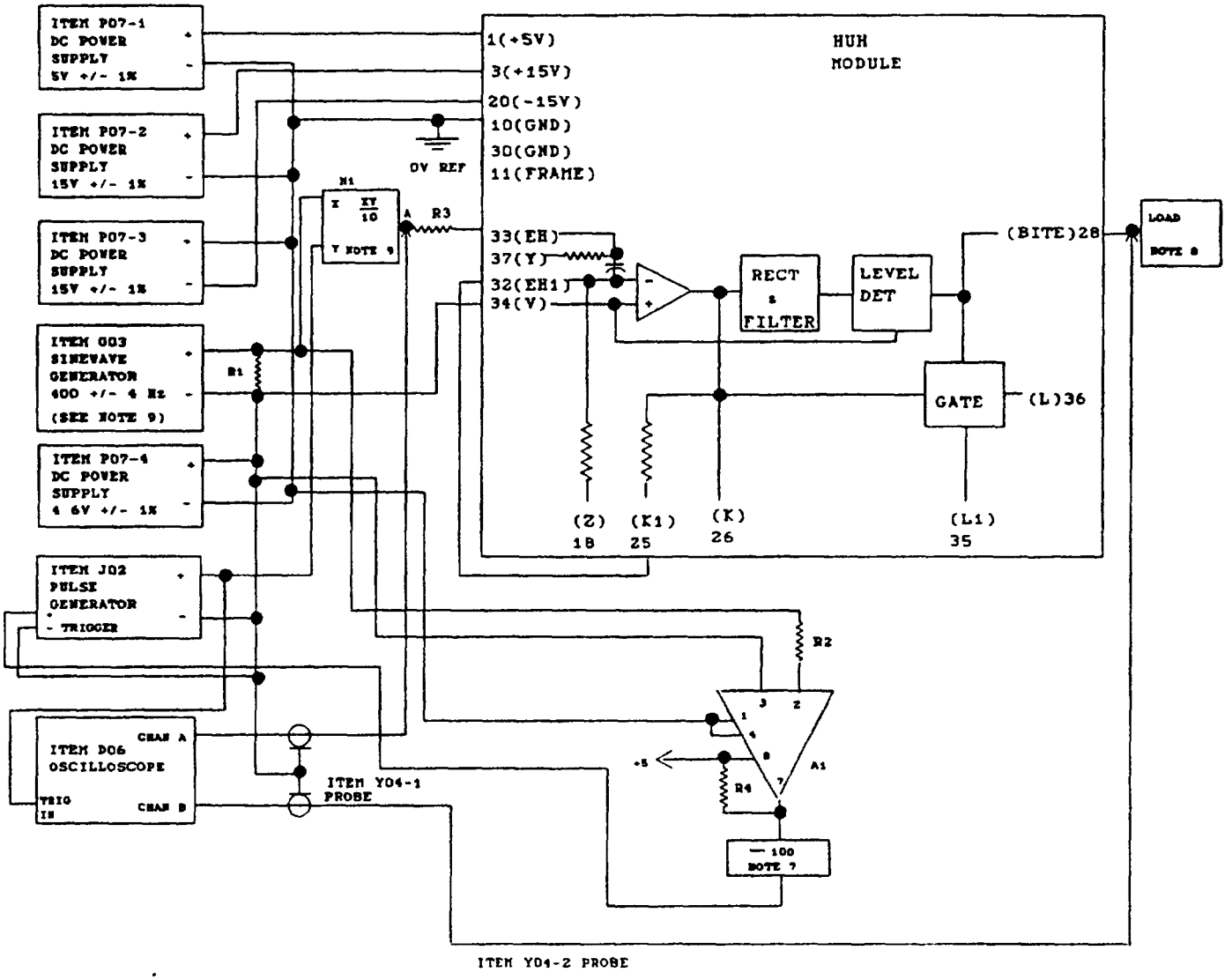
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NOTES:
1. R1 = 90.6 kOhm ± 1%, 1/8 W.
2. R2 = 1 kOhm ± 1%, 1/8 W.
3. R3 = 600 Ohm ± 5%, 1 W.
4. R4 = 10 Ohm ± 1%, 1/2 W.
5. R5 = 12 Ohm ± 10%, 1/8 W.
6. R6 = 20 Ohm, 10 turn trim potentiometer.
7. R7 = 10 Ohm ± 1%, 1/2 W.
8. S1 = S2 = SPST switch.
9. A1 = Fairchild μA791 OpAmp, or equivalent.
10. Connect point A to points B and C (module removed from socket).
11. Adjust R6 to obtain a reading of 500 ± 10 mVpk across R7.
12. Disconnect points A and C, insert module, and proceed with test.
13. Connect test for minimum of 10 seconds.
14. The current sense terminal of A1 shall be connected to its output as shown.
15. No readings are required for test conditions D1-D4.
16. The module passes if it passes the remaining tests.

*FIGURE 8. Output short circuit current check test circuit for table V.

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NOTES:
1. R1 = 600 Ohm ± 5%, 1 W.
2. R2 = 5 kOhm ± 10%, 1/8 W.
3. R3 = 45.3 kOhm ± 1%, 1/8 W.
4. R4 = 3.65 kOhm ± 1%, 1/8 W.
5. A1 = LM111H Voltage Comparator.
6. M1 = Analog Devices AD534LD multiplier, or equivalent.
7. Divide by 100 circuit consists of two 7490 decade counters, or equivalent, and one µA9614 line driver, or equivalent.
8. Load BITE (pin 28) per figure 10, note 6.
9. Using item D06, adjust the amplitude of item G03 to 7 Vpk ± 3% output at point A.

*FIGURE 15. BITE propagation delay time (tPHL and tPLH) and transition time (tTLH and tTHL) test circuit for table XII.

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